Software Design Document (SDD) - EagleNav

Group 5

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Contents

1 Version History

Snapshot	Date
Snapshot 1: Initial version	5/7/2026
Snapshot 2: Added features	5/8/2026
Snapshot 3: Updated system design	5/12/2026
Snapshot 4: Final version	5/14/2026

2 Introduction

2.1 Purpose

This document describes the software design of the EagleNav system. EagleNav is an augmented reality navigation platform. EagleNav aims to remove navigation barriers and create a more inclusive campus experience for all students.

2.2 Intended Audience & Reading Suggestions

This document is intended for students, developers, and instructors.

- Students: sections 2,4
- Developers: section 3,4
- Instructors: sections 2,3,4

2.3 Overview

EagleNav is a campus navigation system design to provide routing with a focus on accesibility, safety, and real-time awareness. The platform integrates multiple data sources including campus map data, routing engines, sensor inputs, and external services. The components work together to generate navigation instructs that adapt to user movement.

3 System Architecture

The system includes a mobile frontend, backend server, and database. It is layered into cloud infrstructure, on-device controllers, and on-device services.

3.1 Architecture Breakdown

The system is divided into the following major layers:

3.2 Cloud infrastructure Layer

The cloud layer is reponsible for storing map data and generating routes.

- Campus map data is stored in Cloudflare R2
- The Valhalla routing engine computes possible routes.
- Updating map data requires uploading a new file and restarting routing service.

3.2.1 Client Controller Layer

The client layer runs on the user's device and manages navigation experience.

- **Location Controller**: Handles GPS input
- **Routing Controller**: Requests routes ffrom cloud service
- **Guidance Controller**: Manages navigation
- **Navigation Voice Controller**: Handles spoken instructions
- **UI Navigation Controller**: Controls navigation states: idle, navigating, and arrived

3.3 Service Layer

The service layer provides reusable system functions.

- **Location Service**: Provides GPS updates
- **Valhalla Service**: Sends routing requests and processes responses
- **Compass Service**: Suplies heading informaiton
- **Text-To-Speech Service**: Generates spoken instructions
- **Haptic Feedback Service**: Provides vibration cues for navigation

3.4 Workflow

1. The user selects a destination in the application
2. The Routing Controller sends a request to Valhalla
3. The cloud service returns a route
4. The Guidance Controller begins navigation
5. The Location Controller updates the user position
6. If user deviates from route, The Routing Controller triggers a reroute
7. Updated routes are recieved
8. The system ends navigation when user arrives/cancels

3.5 Workflow Diagram

User Selects Destination

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Navigation Controller Validates Input

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Routing ControllerRequests Route (Valhalla)

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Route Generated

↓

Guidance Controller Activated

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Sensor Fusion + Map Matching Updates

↓

On Route → Continue Guidance
Off Route → Trigger Reroute

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UI Updates (On Track / Rerouting / Arrived)

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Navigation Ends (Cancel / Arrival)

Figure 1: Compact Navigation Workflow

4 User Interface

Users interact through a mobile app with AR directions and voice guidance. The following list represents the primary pages that users will interact with:

- **Home Screen:** Users will be able to access the navigation screen, settings, and get basic information at a glance.

- **Navigation Screen:** Users will be able to navigate the campus using AR + Map Navigation features.
- **Events Screen:** Allows users to view and save campus events.
- **Settings:** Users will be able to locally change the app's theme, navigation volume, AR settings, etc.

4.1 Screen Components

- **HomeMap Screen:** Provides a search bar, map view, and menu.
- **Guidance Screen**: Includes a turnbanner, progess bar of navigation, and an end/cancel option.

4.2 User Interface Flow Diagram

HomeMap Screen

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Search / Select Destination

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Route Preview Screen

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Start Navigation

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Guidance Screen (Turn-by-Turn Navigation)

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On Route → Continue Navigation
Off Route → Recalculate Route

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Arrival Screen (Destination Reached)

↓

End Navigation

Figure 2: User Interface Flow of EagleNav

5 Glossary

Term	Definition
AR	Augmented Reality
UI	User Interface

6 References

- Ascent - Project (https://ascent.cysun.org/project/project/view/248)